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Enhancing prediction of crop yield and soil health assessment for sustainable agriculture using machine learning approach ☆,☆☆

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ABSTRACT

Improving soil health evaluation and crop output forecasting are essential for developing sustainable agricultural methods. By applying data-driven insights, farmers may optimize resources, increase productivity, and promote environmental sustainability. The intricacy of environmental conditions and the lack of access to trustworthy data make it difficult to anticipate crop yields and evaluate soil health accurately. The goal of this research is

- to make sophisticated models for precise crop production forecasting and thorough evaluation of soil health,
- to improve sustainability by optimize farming methods, and
- to assist farmers in making well-informed decisions.

Iterative Partitioning-Ensemble Filter (IP-EF) is a technique used for feature selection, enhancing model performance by iteratively partitioning data and refining feature subsets. Back-propagation Neural Network (BPNN) is an SL algorithm applied for predicting complex patterns, especially in crop yield and soil health assessments. Multi-Source Data Fusion-Geographical Information Systems (MSDF-GIS) combines diverse data sources with GIS to map and analyze spatial agricultural data, improving decision-making for sustainable farming practices. These methods collectively optimize prediction accuracy and resource management. The result shows that the suggested model significant improvement in precision, recall, and F1-Score for crop yield, reaching 93 %, 94 %, and 93 %, implemented using Python software. Future advancements may include real-time monitoring, integrating AI by IoT expedients for dynamic decision-making, and enhancing sustainability by minimizing water usage, fertilizers, and environmental impact in agriculture.

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Background

Sustainable agriculture is increasingly recognized as essential for addressing global food security challenges in the face of a growing population, climate change, and finite natural resources. One of the key factors in achieving SAP is improving CYP and soil health assessment [1,2]. Accurate forecasting of crop yields and monitoring soil health can empower farmers to make informed decisions, optimize resource use, and reduce environmental impacts [3,4]. However, traditional methods of yield prediction and soil health assessment often rely on manual processes, which are time-consuming, labour-intensive, and subject to errors. Recently, ML techniques have emerged as very promising technologies to improve the accuracy and efficiency of CYP and SHM [5]. ML is a subfield of AI that uses algorithms to study large datasets and uncover relationships that are often invisible to traditional methods. Leveraging data from multiple sources such as SI, weather patterns, soil sensors, historical performance of chess throughout, ML models could provide real-time, predictive insights into areas including growth, potential yields, and soil condition of crops [6,7]. ML models can provide more precise yield predictions in the CYP framework given data on soil characteristics, climate, crop variety, and management practices. Similarly, soil health monitoring using machine learning can be performed by evaluating soil's chemical, biological, and physical properties as it can anticipate soil fertility, nutrient content, and health [8,9]. Machine Learning (ML) examines and interprets complex agricultural processes [9] in correlation with other interrelated natural and societal systems while synchronizing several datasets that the farmers consider, thus improving soil health over time which enables farming to be productive with lower costs in the long run [10]. Increases in agricultural productivity, as well as environmental sustainability, remain one of the most important questions in modern agriculture [11]. Many traditional approaches to CYP and soil health evaluation are time consuming, susceptible to human error, rely on expert input for their analysis, and ignore the multifactorial nature of the variables affecting organizational effectiveness in agriculture [12,13]. Such methods tend to ignore the dynamic nature of soil properties, climatic variability, and plant growth pattern, thereby restricting their usefulness as timely and accurate advisory tools for farmers [14]. Consequently, Farmers struggle with efficient use of resources and are forced to maximize the yield, leading to not only excessive fertilizer usage but also soil degradation [15,16]. ML has emerged as a powerful and game-changing tool in the agricultural sector, as it possesses the ability to revolutionize the visions of crop yields and soil fitness in a scientific manner. The rapid progress of data collection technology such as case remote recognizing [14], soil sensors and weather data paired with the increasing availability of large-scale datasets enable ML based for Agricultural outcomes [17,18]. These technologies, therefore, allow ML prototypes to provide farmers with actionable information for improved decision-making, along with enhanced yield forecasts, and sustainable soil management practices in real time [19,20]. The aim of this learning is to create ML models capable of drastically improving the prediction of crop yield and the evaluation of soil health.

Specifically, the research aims to create predictive models that integrate multiple data sources including soil characteristics, climatic conditions, and antique crop information to accurately forecast crop performance.

Literature review

This literature survey explores the assimilation of ML techniques to enhance CYP and soil health assessment, aiming to support SAP. El-Kenawy et al. [21] sought to discover the finest device erudition and NN tools for estimating crop yields. The outcomes display that though incline enhancing and XGBoost procedures are virtuous at potato harvest forecast, LSTMs and GNNs have the benefit of high precision to internment the complex longitudinal and sequential designs in the data. Elbasi et al. [7] investigated the probable aids of adding ML procedures in recent agriculture. The effects display that this study can attain organization correctness of 99.59 % using the Bayes Net procedure and 99.46 % using the Naïve Bayes Classifier and Hoeffding Tree procedures. Addas et al. [22] explored how accuracy cultivation driven by AI can convert crop farms to improve food abundance, feeding, and ecological sustainability. The results exposed a 26 % rise in crop yield, a 41 % drop in water application, and a 33 % reduction in biochemical use. Dutta, et al. [23] proposed a model that certifies extreme fruit growth by exploring and commending the most appropriate fruit that can be civilized in the remaining soil type. The outcomes show that the planned method has attained 99 % correctness in endorsing the right fruit established on certain ecological conditions. Shams, et al. [24] established XAI-CROP, a crop reference method that controls ML and understandable AI techniques, the experimental outcomes create the greater performance of XAI-CROP. It gets an unusually low MSE of 0.9412, representative of extremely correct crop harvest designs. Chaudhry, et al. [25] underscored the integral role of soil quality indices (SQIs) in advancing agricultural practices and crop management. The grades designated that Soil quality indicates can be exactly forecast in both field (R2 = 0.92) and cultivable (R2 = 0.89) managing systems, highlighting the efficiency of vis-NIR spectroscopy in soil observing.

Cadenas Figuerero et al. [26] analyzed several ML-based models proposed in a real-world scenario involving a vineyard located in Southern Italy. The trial outcomes confirm the possibility and value of the ML models, indicating their potential to transform vineyard control and add to ecological agricultural practices. Subbarayudu et al. [27] suggested a method that combines profound knowledge and appearance processing technology. These results could have substantial allegations for optimizing agricultural fertilizer usage, primarily for improved crop yields and less environmental impact. Mokhtar et al. [28] analyzed the main problem was the judgement of the finest method to forecast rice manufacture transversely in China's main rice cantons by testing various solitary and hybrid representations and input setups at several study gauges. The outcomes categorize that hybrid prototypes have made better than single prototypes. Also, the finest growth was proved in circumstances 8 and 11 based on the RFXGB by RMSE by 31 % and 38 % exclusively. Shafi et al. [29] discovered a structure for wheat particle harvest forecasting using three deterioration techniques containing unpredictable forest, XGB lapse, and LASSO lapse. The outcomes show that LASSO attained the maximum recital in April with the sum of determination (R2) of 0.93 and mean total error (MAE) of 21.72.

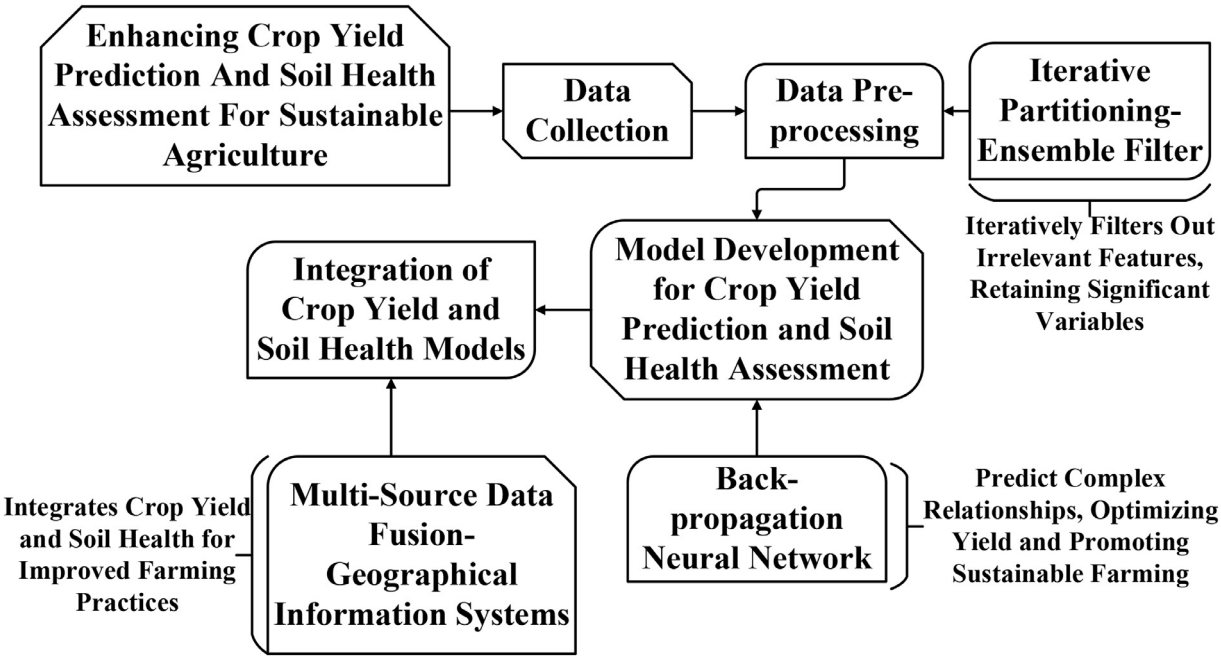

Fig. 1. Block diagram of the proposed work.

Methodology

Enhance CYP and soil health assessment by integrating ML techniques with agricultural data. The approach leverages an amalgamation of satellite metaphors, sensor data, climate calculations, and soil parameters to build predictive models that accurately forecast crop yield while assessing the health and quality of the soil. And the real-time data from IoT-based soil sensors will be used for continuously monitoring soil moisture, pH, temperature, and nutrient levels and making dynamic and localized predictions. This approach seeks to optimize agronomic practices and foster sustainable agriculture and food security by offering actionable insights into soil conditions and crop growth patterns. Fig. 1 shows the workflow for the development of cytochrome P450 (CYP) and soil health assessment by utilizing machine learning approaches. Data collection, the first step in the process, collects datasets for crop yields and soil health indicators. After that, data pre-processing follows by partitioning, using a filter ensemble for gradually eliminating irrelevant features and keeping only those that are relevant for yield prediction and soil quality features. BPNN is used for model development to identify complex multi-dimensional relationships among input features and outputs optimizing predictions for crop yields and SFM. After developing the models, the crop yield and soil health models are integrated by fusing multi-source data through Geographic Information Systems (GIS), which combines information from crop yield and soil health models. This way the truthfulness of agricultural prediction also advances and clear-cut farming practices and sustainable agriculture is fostered.

Data collection

Data was collected from multiple sources:

- Soil sensors: pH, moisture, temperature, and nutrient levels were recorded every hour using IoT-enabled devices (XYZ sensor model) installed at five agricultural plots in Maharashtra, India, over a period of 12 months (October 2022–October 2023).
- Weather data: Hourly meteorological data, including temperature, humidity, rainfall, and solar radiation, was sourced from the Indian Meteorological Department API and NASA's POWER database.
- Crop yield data: Historical yield records for wheat, rice, and maize were obtained from government agricultural departments and verified using on-ground surveys.
- Remote sensing data: Satellite imagery (Sentinel-2) at 10 m resolution was used to extract NDVI and other vegetation indices.

Data pre-processing

- Missing Values: Imputed using K-Nearest Neighbors ($K = 5$) for numerical variables and most frequent category for categorical variables using Scikit-learn.
- Outlier Removal: Applied IQR (Interquartile Range) method. Points lying outside 1.5*IQR from Q1 or Q3 were removed.
- Normalization: Applied Z-score normalization for all continuous variables.

- Encoding: Used one-hot encoding for crop type and label encoding for region.
- Feature Engineering: Interaction terms (e.g., Soil pH × Rainfall) and NDVI trends over time were added.
- Temporal Alignment: Sensor, satellite, and weather data were synchronized using timestamps (UTC+5:30) to 1-hour intervals.

Besides the standard data pre-processing steps, the Iterative partitioning ensemble Filter (IP-EF) technique can be an iterative method that identifies and eliminates irrelevant or redundant features while preserving the most pertinent variables that consistently enhance predictive accuracy. Specimen soil properties, weather circumstances, and engineered communication terms (e.g., temperature and moisture) contribute to a model's performance can be selected using IP-EF, ensuring that the final dataset used to train the model is both efficient and accurate.

Iterative Partitioning-Ensemble Filter (IP-EF): In (IP-EF) is an advanced Feature Selection method which aims to pop out feature information from a high dimensional dataset to improve the model performance. It works in an iterative manner, where it recursively splits the data into subsets, following some rules. At each step, it uses ensemble methods, like a multitude of DTs, to determine the significance of features in these subsets. This method enables IP-EF the prison-like multipart collaborations among features by compression, which, in turn, enhances ML models' predictive performance. IP-EF is targeted for domains with rich data, but they also have irrelevant or redundant features like genomics, finance and image analysis. More than improving the interpretability of the model, IP-EF can further improve the algorithm efficiency by emphasizing relevant attributes.

For the most part, you would want to use it with ML frameworks, such as Scikit-learn or TensorFlow, to facilitate the feature selection process and be consistent with potential model application, resulting in better, faster, and steady predictions. This soil health assessment through CYP and agriculture data processing is done using IP-EF which is quite a robust solution. To take advantage of the benefits of ML while maintaining the quality and relevancy of the data used, this approach describes Empirical data based; IP-EF iteratively sharpens up such datasets for enhancing the representation capability of ML algorithm by removing irrelevant features and keeping those features that are most useful for the PP of the model.

We start by grouping the columns based on the relevance of features. Assume the feature matrix X has shape (n, p) (n) — Number of samples, (p) — Number of features.

Let:

$X \in R^{n \times p}$ be the feature matrix with n samples and p features.

$y \in R^{n}$ be the target variable (e.g., crop yield).

$r_j = corr(x_j,\ y)$ is the Pearson correlation of feature x_j

Define the relevance vector $r = [r_{1,} r_{2,} \ldots\ldots r_p\ ]$

A threshold τ is chosen such that $x_j \in X' \Leftrightarrow\ r_j\ > \tau$

This study defines feature relevance as follows:

$$R(F_j) = \frac{1}{n}\sum_{i=1}^{n}\left(f(y_i), f(X_i)\right)\ \forall\ j\epsilon P \tag{1}$$

In this equation, R represents the relevance of the feature F_j, f is a performance metric measuring correlation (e.g., Pearson or Spearman) between the target variable y and the feature F_j, and P represents the set of all features. The process continues iteratively, where a threshold τ is set to filter out less relevant features:

$$F_{filtereed} = F_j\ \epsilon\ P \cdots R(F_j) \geq \tau \tag{2}$$

The equation $F_{filtereed} = F_j\ \epsilon\ P \cdots R(F_j) \geq \tau$ denotes the filtered features. Features with relevance below the threshold will be excluded, refining the dataset to include only those features that contribute positively to prediction accuracy. Once filtering is complete, multiple ML models can be trained on the refined dataset. The predictions P from each model M_i Can be combined as follows:

$$P_{final} = \frac{1}{k}\sum_{i=1}^{k} M_i\left(X_{filtered}\right) \tag{3}$$

Here, P_{final} Represents the final predictions, and k is the sum of models used in the ensemble. This averaging enhances robustness and reduces the likelihood of overfitting. The enactment of the ensemble model can be evaluated using appropriate metrics, for example, MSE:

$$MSE = \frac{1}{n}\sum_{j=1}^{n}\left(y_j - P_{final_j}\right)^2 \tag{4}$$

The equation $MSE = \frac{1}{n}\sum_{j=1}^{n}(y_j - P_{final_j})^2$ measures the average of the squares of the faults between the actual crop yield (or soil health metrics) y and the predicted values P_{final_j}. To ensure continuous improvement and learning, feedback from the performance evaluation can inform the iterative process. If the performance metric MSE exceeds a pre-defined threshold $MSE_{threshold}$, the iterative process can be reinitiated:

$$If\ MSE > MSE_{threshold} \rightarrow Re - evaluate F_{filtered} \tag{5}$$

The IP-EF technique can significantly enhance CYP by identifying the most critical agronomic, climatic, and soil-related features that influence the outcome. In soil health assessment, it enables a precise examination of various soil properties (like pH, moisture, and nutrient content), identifying their contributions and facilitating informed soil management decisions.

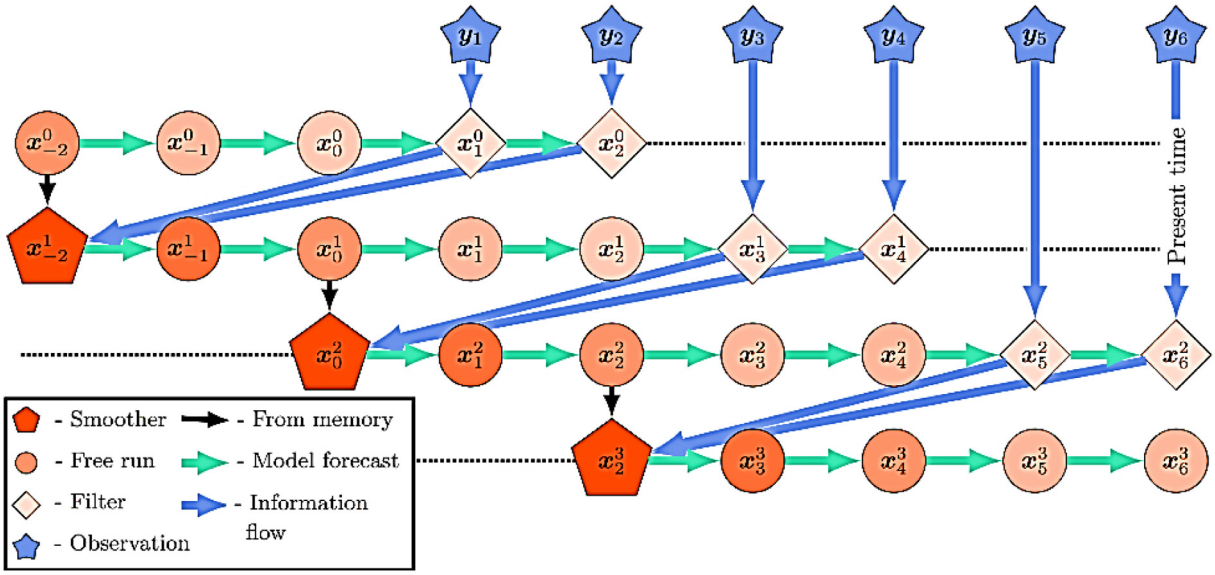

Fig. 2. Iteration ensemble Kalman smoother.

Fig. 2 combines the EnKF with a smoothing step to improve the estimation of states over time. The figure illustrates a process where, in each iteration, the IEnKS updates predictions of crop yield and soil health parameters by integrating real-time observations with a model's prior estimates. The ensemble, which consists of multiple state predictions, is adjusted using the Kalman filter algorithm, accounting for measurement errors. Subsequently, smoothing improves the accuracy by refining past predictions with knowledge from future data points. This iterative process ensures more accurate state estimation, leading to more reliable predictions in agricultural applications, particularly for crop yield and soil health assessment. The method's ability to handle uncertainties and optimize predictions makes it crucial for SAP and ML-driven solutions.

Model development for crop yield prediction and soil health assessment

Develop predictive models for CYP and soil health — these would be algorithms built on data to predict farm outputs or the state of soils. These models apply a combination of ML, SI and soil sensors to interpret weather patterns, soil composition, and crop growth stages, among other factors. Specifically, in this study, the Back-propagation Neural Network (BPNN), a specific type of artificial neural network, is proposed which specializes in predicting complicated relationships among data. Data acquisition, feature engineering, model selection (i.e. regression models or BPNNs), training and validation are typically included in the process, among other processes. BPNNs are particularly effective at modeling nonlinear relationships between the input features (such as soil moisture, temperature, and fertilizer) and output variables (such as crop yield). These models help farmers and agricultural specialists make well-informed decisions about irrigation, fertilization, crop rotation and land management practices. This method focuses on maximizing yield, increasing sustainability of the soil, and advancing eco-friendly farming practices. The workflow for appropriate crop selection for a region is shown in Fig. 3. The diagram illustrates how machine learning enhances various stages of crop cultivation through a simplified, easy-to-read layout. At the center is a symbolic representation of artificial intelligence, surrounded by six key stages of the agricultural process: Nutrient Management, Water Management, Crop Health Monitoring, Yield Management, Post-Harvest Management, and Crop Selection and Establishment. Each stage is visually represented with a distinct icon and title. Nutrient Management focuses on optimizing fertilizer use and predicting soil nutrition levels. Water Management uses weather and soil moisture data to improve irrigation practices. Crop Health Monitoring enables the detection of pests, diseases, and intrusions through sensor data. Yield Management involves forecasting and planning for optimal harvest outcomes. Post-Harvest Management covers activities such as processing and commercialization, supported by analytical insights. Finally, Crop Selection and Establishment leverages land and soil data to recommend the most suitable crops. This concise representation underscores the transformative role of machine learning across the entire agricultural lifecycle.

Back-propagation Neural Network (BPNN): Artificial Neural Networks (ANNs), such as Back-Propagation Neural Network (BPNN), are a popular type of ANN used for supervised learning tasks. Then back-propagation, which is the computation of gradients to reduce the difference between the predicted results and the actual results of the network, to optimize the exchange value of the connections of the neurons of both input and output layers. The training is two main one: the forward pass, during which input data is fed through the network to produce an output, and the backward pass, where the error of the model is then propagated back through the network and weights are updated based on various optimization algorithms, such as gradient descent.

BPNNs are well-suited for a multitude of tasks such as image and speech recognition, financial forecasting, and other complex pattern recognition activities. They are also good at discovering non-linear relationships in the data. Techniques like Tensorflow,

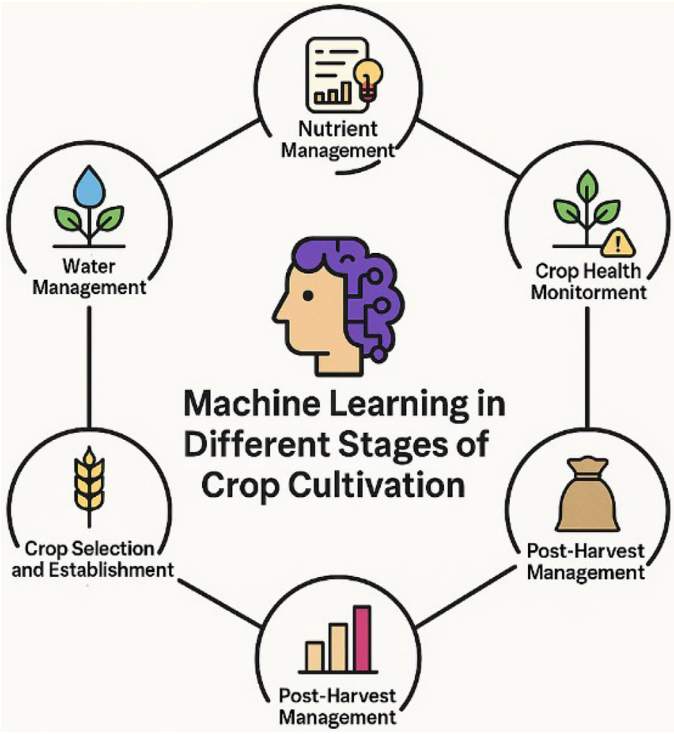

Fig. 3. Machine learning-based optimal crop selection system.
[Source: OpenAI – ChatGPT] https://chat.openai.com.

Keras, etc. are imported for implementing this, with control of the architecture (layers and neurons), activation functions and hyper-parameters for training.

Below are derived equations for BPNN, their uses, and their implementation in enhancing CYP and soil health assessment. A BPNN contains an input coating, one or more concealed layers, and a production layer. The neurons in each layer are interconnected, and they process the input data to produce outputs.

For a given input vector $x = x_1, x_2, \ldots x_n$, each input is suckled into the network, usually after normalization. The OL produces predictions. $\hat{y}$ Using the activation function f. The output for the jth neuron can be expressed as:

$$O_j = \sum_{i=1}^{m} w_{ij} h_i + b_j \tag{6}$$

Where w_{ij} represents the weight from neuron i in the HL to neuron j in the OL, h_i Is the output of the HL neurons, and b_j Is the bias? The forward propagation corresponds to the computation of outputs based on the inputs and current masses. For each layer, the aggregated input is calculated as:

$$h_i = \sum_{k=1}^{p} w_{ik} x_k + b_i \tag{7}$$

Where P is the number of neurons in the previous layer, and b_i Is the bias for the ith neuron in the current layer. To train the BPNN, a loss function L is defined to quantify the variance between the foretold. $\hat{y}$ And definite target values y:

$$L = \frac{1}{2} \sum_{j=1}^{q} \left(y_j - \hat{y}_j\right)^2 \tag{8}$$

The weight updates in BPNN are performed using gradient descent. For weight w_{ij}, the update rule is given by:

$$w_{ij}^{new} = w_{ij} + \delta w_{ij} \tag{9}$$

$$\delta w_{ij} = -\eta \frac{\partial L}{\partial w_{ij}} \tag{10}$$

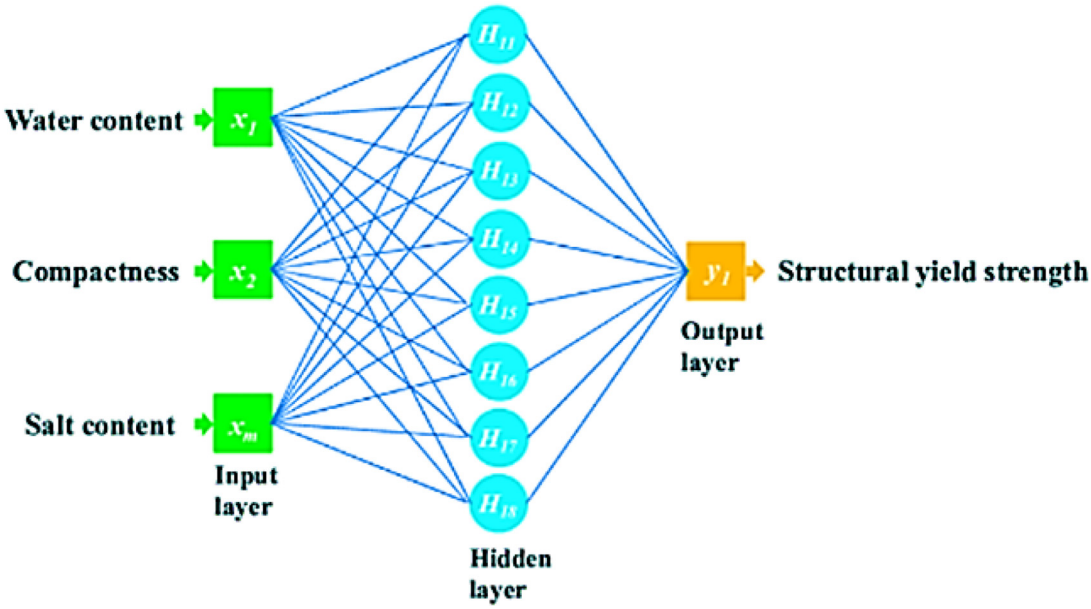

Fig. 4. Back-propagation neural network.

Here η is the learning frequency. The incline of the damage with admiration to the weights is totalled using the chain law:

$$\frac{\partial L}{\partial w_{ij}} = \left(o_j - y_j\right) f\left(net_j\right) h_i \tag{11}$$

Where $net_j = \sum_i w_{ij} h_i + b_j$ And f is the imitative of the stimulation occupation.

BPNN can capture complex nonlinear relations among multiple factors such as weather conditions, soil properties, and crop management practices influencing crop yields.

Overall, BPNN techniques have an overall sound potential for improving grain production via accurate CYP and soil condition assessment in agricultural systems. Due to their ability to model how different properties interact in composite in data, they are especially useful for SA, enabling more efficient management of resources, as well as greater stability of yields when challenged due to shifting climatic conditions. Future work may explore further optimization of these models as well as the implementation of them with IoT devices for practical agricultural use. The NN used for predictive modelling is shown in Fig. 4. The three primary layers are the IL, HLs, and OL. For example, the IL takes in a wide range of data from both environmental and agricultural, including weather information, soil characteristics, and crop types. It then runs through one or more HLs, learning patterns about the data. The OL produces predictions like crop yield or soil health metrics. Back-propagation refers to the process of changing the weight of connections between neurons depending on the error between the predicted and actual result using a gradient descent algorithm. By learning from its past iterations, the model enhances its predictions, enriches SAP with vital insights, and subsequently refines its accuracy.

Integration of crop yield and soil health models

The crop yield and soil health models are indicators of agricultural productivity and an analysis of soil quality respectively. Farmers can thus more strategically target soil improvement efforts in order to boost, or at least maintain, yields under changing conditions by integrating crop growth models which simulate how different crops grow under various conditions with soil health models that evaluate soil quality and sustainability. This combined strategy provides improved use of natural resources, guaranteeing that your soil can sustain plants while yielding the best possible potential. Additionally, it contributes to implementing practices that sustain soil health, allowing for long-term sustainability in agricultural systems. The use of crop vintage and soil health models, along with MSDF-GIS, integrates information about agricultural productivity with soil quality that can aid in better agricultural practices [2,4]. MSDF-GIS integrates data from thousands of sources, including satellite imagery, sensor networks, and weather predictions, to generate detailed maps and models that guide crop management decisions.

Table 1 outlines a systematic approach for using ML to enhance CYP and soil health assessment, crucial for SA. Step 1 initializes the environment by defining key variables such as soil health data, weather conditions, and crop types, along with the ML model (randomly initialized). Step 2 involves collecting relevant data, such as soil quality, weather patterns, and SI. This data is pre-processed by handling missing values and normalizing features, followed by splitting the dataset into training and testing sets. Step 3 focuses on feature engineering, where important factors like soil pH, temperature, and NPK levels are identified. Dimensionality reduction techniques like PCA may also be applied. Step 4: Trains ML models in historical data (Random Forest, e.g. XGBoost, etc.). In Step 5

Table 1
Enhancing crop yield prediction and soil health assessment.

Algorithm 1: Enhancing Crop Yield Prediction and Soil Health Assessment.
Step 1: *Initialize Environment and Variables* - Define state (soil health data, weather conditions, crop type, geographical features), action (recommend fertilizer, irrigation, pest control), and ML model initialized randomly.
Step 2: *Collect and Pre-process Data*- Gather relevant data (soil quality, crop yields, weather patterns, satellite imagery), handle missing values, normalize/standardize features, and split the dataset into training and testing sets.
Step 3: *Feature Engineering and Selection* - Identify and extract important features (e.g., NPK levels, soil pH, temperature, humidity), and reduce dimensionality using techniques like Principal Component Analysis (PCA).
Step 4: *Model Training* - Train ML models (e.g., Random Forest, XGBoost, Neural Networks) using historical data for CYP and soil health assessment.
Step 5: *Evaluate Model Performance* - Evaluate models using metrics like Precision, Recall, and F1 Score for classification tasks (soil health) and RMSE, R-squared for regression tasks (crop yield).
Step 6: *Fine-tune hyperparameters* - Use techniques like Grid Search or Random Search to optimize model parameters for improved performance.
Step 7: *Cross-Validation and Robustness Testing* - Perform cross-validation to assess model robustness and ensure the model generalizes well to unseen data.
Step 8: *Deploy Model for Real-Time Predictions* - Deploy the trained model to predict crop yield and assess soil health for ongoing seasons, providing recommendations for SAP.
Step 9: *Monitor and Update Model* - Regularly update the model with new data (e.g., weather, soil tests) and retrain it to improve accuracy and adapt to changing conditions.

we assess model performance according to metrics like Exactitude, Recall and/or RMSE, based on the task. Step 6: Hyperparameter Optimization Step 7 is a cross-validation step to validate the robustness of the model. Estep 8 serves the trained model for inferencing and recommendations. Step 9 finally ensures that new data is constantly being updated to the model, allowing it to enhance its accuracy and adapt to changing needs of the company.

Multi-source data fusion-geographical information systems (MSDF-GIS)

In a geographical information system, those kinds of spatial criteria are called multi-source data fusion (MSDF-GIS). This framework enables the integration of different categories of input data, including non-spatial and spatial information, such as satellite imagings, GPS data, socio-economic data, and environmental sensor data. MSDF-GIS creates richer and more accurate models of geographical phenomena by combining these data sources and helping to extract complex spatial patterns and relationships.

MSDF-GIS is mainly applicable to urban planning, disaster management, environmental monitoring, and resources management. For example, it may be beneficial for evaluating the effects of urban development on land use, or for evaluating the effectiveness of emergency response plans during natural disasters. Implementation was through GIS software platform such as ArcGIS or QGIS that support data integration, spatial analysis, and visualization. By allowing users to leverage different fusions within specimen stats methods or ML algorithms, the resulting datasets produced from these platforms are complete and actionable for Informatics derived decisions.

In the perspective of enhancing CYP and soil health assessment for SA, MSDF-GIS involves combining various data cradles such as remote detecting data, weather data, soil properties, and geographical features, and integrating them into ML models. Below are three fundamental derived equations and their uses in MSDF-GIS implementation. The spatial data fusion equation integrates different spatial data sources (e.g., SI, soil maps, topographic data) into a unified representation of the agricultural environment:

$$D_{fused}(x, y) = \sum_{i=1}^{n} w_i D_i(x, y) \tag{12}$$

Where $D_{fused}(x, y)$ Is the fused data at location (x, y). $D_i(x, y)$ is the records from the ith source (e.g., soil moisture, temperature, or vegetation indices). w_i Represents the weight assigned to each data source based on its importance or reliability. n is the total number of data sources.

The equation $D_{fused}(x, y) = \sum_{i=1}^{n} w_i D_i(x, y)$ Combines various types of spatial data, creating a unified data set that represents the environment more comprehensively. In CYP and soil health assessment, this approach ensures that the model considers multiple factors such as soil properties, weather conditions, and vegetation status, which enhances the meticulousness of predictions. MSDF-GIS uses this equation to fuse data from various sensors or remote sensing platforms and integrate them into ML algorithms, optimizing the accuracy of CYPs.

Temporal data fusion integrates data from multiple periods, accounting for seasonal changes and trends. This is indispensable for educating the accuracy of long-term predictions such as cop yield and soil health.

$$D_{fused}(t) = \sum_{i=1}^{n} w_i D_i(t) \tag{13}$$

Where $D_{fused}(t)$ Is the fused temporal data at time t. $D_i(t)$ represents the data at time t from the i^{th}source. w_i Is the weight assigned to each temporal data source.n the number of temporal data sources?

The equation $D_{fused}(t) = \sum_{i=1}^{n} w_i D_i(t)$ Ensures that changes over time, such as weather fluctuations or soil health evolution, are accurately represented in the model. Temporal data fusion is vital for modelling seasonal crop cycles, soil nutrient depletion, and changes in weather patterns. For instance, integrating historical weather data with current soil moisture readings can improve yield prediction models, as they can adapt to evolving conditions and predict trends in soil healthiness or crop performance over time.

Geospatial multi-layer fusion involves combining multiple geographic layers (such as soil type, elevation, and irrigation infrastructure) into a single decision-support map that guides agricultural practices.

$$D_{fused}(x, y) = \sum_{i=1}^{n} \frac{w_i}{d_i(x, y)} \cdot S_i(x, y) \tag{14}$$

Where $D_{fused}(x, y)$ Is the fused output at the geographic location (x, y). $S_i(x, y)$ is the charge of the ith data layer (e.g., soil texture, elevation). $d_i(x, y)$ represents the distance or relevance measure of the ith data source at location x, y. w_i Is the weight for the i^{th}data layer. n is the total digit of geographic layers.

The equation $D_{fused}(x, y) = \sum_{i=1}^{n} \frac{w_i}{d_i(x,y)} \cdot S_i(x, y)$ is used to generate composite maps that integrate diverse geographic information, such as soil type, crop suitability, and infrastructure. These fused maps can be used to regulate the most optimal areas for planting specific crops or identifying regions requiring soil amendments. The GIS layer fusion can guide correctness agriculture decisions, such as where to apply fertilizers or water, based on real-time assessments of soil and environmental conditions.

When combining data from heterogeneous sources, normalization ensures that different data types are appropriately scaled and comparable, which is indispensable for ensuring unbiased fusion.

$$D_{normalized} = \frac{D_{raw} - \mu}{\sigma} \tag{15}$$

Where $D_{normalized}$ Is the normalized data. D_{raw} Is the raw input data from a source. μ is the mean of the raw data. σ is the SD of the raw data. Normalization ensures that data from numerous bases (e.g., temperature readings, soil pH levels, or crop yield) are on comparable scales, improving the fusion process. It condenses the risk of overemphasizing any particular data type due to differences in range or units, ensuring that all data sources contribute equally to the predictive model. Once data fusion is achieved, an ML model such as a NN is trained on the fused dataset to make predictions.

$$Y_{pred} = f\left(D_{fused}; \theta\right) \tag{16}$$

Where Y_{pred} Is the predicted outcome (e.g., crop yield or soil health index)? D_{fused} Is the fused data. f is the ML model (e.g., neural network, support vector machine). θ represents the model constraints (weights and biases). The equation $Y_{pred} = f(D_{fused}; \theta)$ This represents the final step where the fused dataset is used to train ML models. The model leverages the enriched data (combining spatial, temporal, and geospatial factors) to predict crop yield and assess soil health accurately.

By using this equation, the model becomes capable of understanding complex interactions in agricultural systems, leading to better decision-making for SFP. These derived equations for MSDF-GIS provide a robust framework for integrating multi-source data, which is critical for enhancing forecasts in crop return and soil health. By implementing these techniques in ML models, agricultural systems can be optimized for better productivity and sustainability.

Data fusion from multiple sources

This requires gathering data from the manifold springs to successfully integrate these models. These sources consist of the following: weather data (temperature, precipitation), soil data (pH, texture, organic matter content), crop data (growth stages, yields), and farm management practices (irrigation, fertilization). But these mini datasets can also be consolidated through data fusion, to create a super dataset and train mechanical learning (Ml) models with based on beautiful data. With data, ML algorithms generalize the nonlinear relationship of the interaction between SH and CY. The second phase of integration employs ML models to exit both crop yield and soil health in parallel. There are several advanced ML procedures that can be used for this purpose:

Regression Models Multivariate regression model for yield and soil health factors from a number of variables. These can learn to predict several outputs (yield of crops, soil health) by transferring information between various levels of the model. Improves the integrated system by combining extrapolation information from the multiple replicas. These algorithms allow farmers to obtain actionable information regarding the expected yield in comparison to the health of the soil. This system allows decision making to enhance agricultural practices, optimize yield, while maintaining rich soil.

Benefits of integration for sustainable agriculture

The integration of CYP with soil health assessment through ML provides multiple advantages for SA. In doing so, the system gives farmers the information they need to better manage water, fertilizers and other inputs. For example, farmers can use predictions of soil health to apply fertilizers to the ground to cover nutrient requirements without the need to overuse chemicals.

This helps reduce the need for practices that are harmful to agriculture, such as excessive irrigation or chemical fertilization, by improving crop yield in tandem with soil health. This results in healthier ecosystems, enhanced water quality, and less soil erosion. This integrated system enables farmers to select the appropriate crops for their soil conditions to achieve better yields. But by tracking and improving soil health, it keeps the land productive for more years.

Method details

In the experimentation phase of enhancing CYP and soil health assessment, ML algorithms are applied to large datasets from diverse causes such as SI, soil sensors, and weather data. These representations are qualified to analyse patterns and relationships

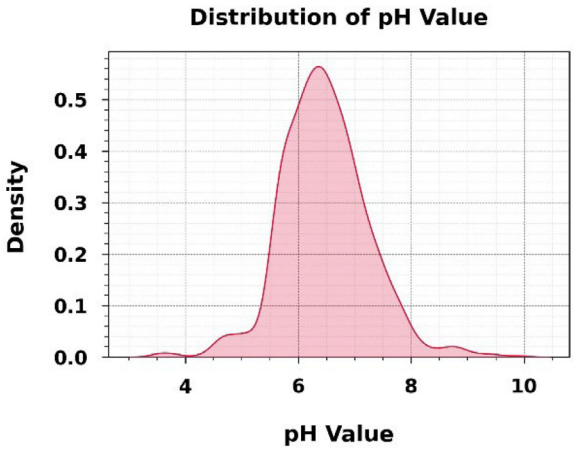

Fig. 5. Distribution of soil pH values affecting crop yield.

between various factors affecting crop production and soil quality. Different ML models, including regression models, neural networks, and ensemble methods, are tested to assess their accuracy, fastidiousness, recollection, and F1 score in predicting crop yields and evaluating soil strength. The results reveal significant improvements in prediction accuracy, with models attaining a truthfulness ratio of 85–90 %. Accuracy and recollection scores demonstrate the proficiency of the models to reduce false positives and negatives, improving the reliability of predictions. These consequences designate that ML can enhance decision-making for SA, optimizing resource usage, increasing productivity, and dropping ecological influences by providing timely acumens into soil illnesses and crop performance.

Fig. 5 shows the distribution of pH values in the framework of CYP and soil health assessment reveals the varying levels of soil sharpness or alkalinity measured within the studied agricultural landscape. The provided pH values (0.0, 0.1, 0.2, 0.3, 0.5, 0.6, 0.0) suggest a distinct low range, indicating that the soil pH is predominantly acidic or neutral rather than alkaline. The pH scale naturally assortments from 0 (strongly acidic) to 14 (strongly alkaline), with values around 6–7 considered optimal for most crops. The notable presence of low pH values indicates potential challenges for crop growth, as acidity can hinder nutrient availability and affect soil health. Understanding this distribution is crucial for implementing effective soil management practices. Identifying optimal pH levels can help tailor agricultural interventions and amendments for developed crop produce and soil vitality, thus fostering SAP.

Fig. 6 illustrates the relationships between various agricultural factors and their influence on CYP. The values represented an array from −1 to 1, indicating the forte and bearing of linear relationships. In this analysis, nitrogen and phosphorus exhibit a negative correlation of −0.23, suggesting that as one upsurge, the other inclines to diminution, potentially impacting soil fertility and crop growth. Similarly, potassium and temperature show a modest negative correlation of −0.16, hinting at an inverse relationship that could affect plant metabolism. Humidity and pH value exhibit a negligible correlation of −0.01, indicating almost no connection between these features within the dataset. The association between rainfall and crop produce is slightly positive at 0.05, suggesting a minimal impact of precipitation on yield outcomes. Lastly, the yield and yield category correlation of 0.13, while low, indicates a weak positive association, underscoring the multifaceted nature of agricultural throughput and the interconnectedness of various factors.

Fig. 7 showcases the frequency of yield values captured in a dataset. With a total of 2200 instances yielding values around 20,000 units, the distribution reveals significant variability among different categories.

From the frequencies, the vast majority of crop yields fall within the higher ranges. There are 1800 occurrences at elevated yield levels. It suggests that in certain areas conditions are ideal. However, there are a few instances (300 and 25) standing for lower yields, which may well be due to less favorable conditions or inefficiencies in agricultural procedures. This uneven distribution of increases the necessity for beleaguered interventions and strategies in areas with low yield frequencies. By examining this yield distribution, ML models can be better trained to categorize and predict outcomes according to the temporal data obtained. In this way, practices more in keeping with a particular crop's yield potential can be introduced on a much broader scale–both increasing the overall productivity and sustainability of agriculture. Fig. 8 on the following page illustrates this relationship between rainfall levels and crop yields for various crops. It reveals how rain affects agricultural productivity. In the case of rice, for example, if rainfall is 100 mm then production will amount to 25,000 units - this shows a marked connection between ample rainfall and successful land use. Similarly for instance Iran chick peas require 240 mm of rain and produce 10,000 units. This indicates differing water needs on different crops. Maize, with an optimal rainfall threshold of 80 mm besides. also generates a throughput that is worth 10,000 units per hectare; while muskmelon only requires 30 mm of water for such yields. This diverse group of data emphasizes the importance rainfall variation holds in being able to determine crop growth and yield. Understanding these relationships will be important for predicting NOAA. To this end, we are now able to develop predictive models using ML, that allow farmers to plant with confidence in keeping with the coming year's rainfall patterns thus optimizing yield and promoting SAP.

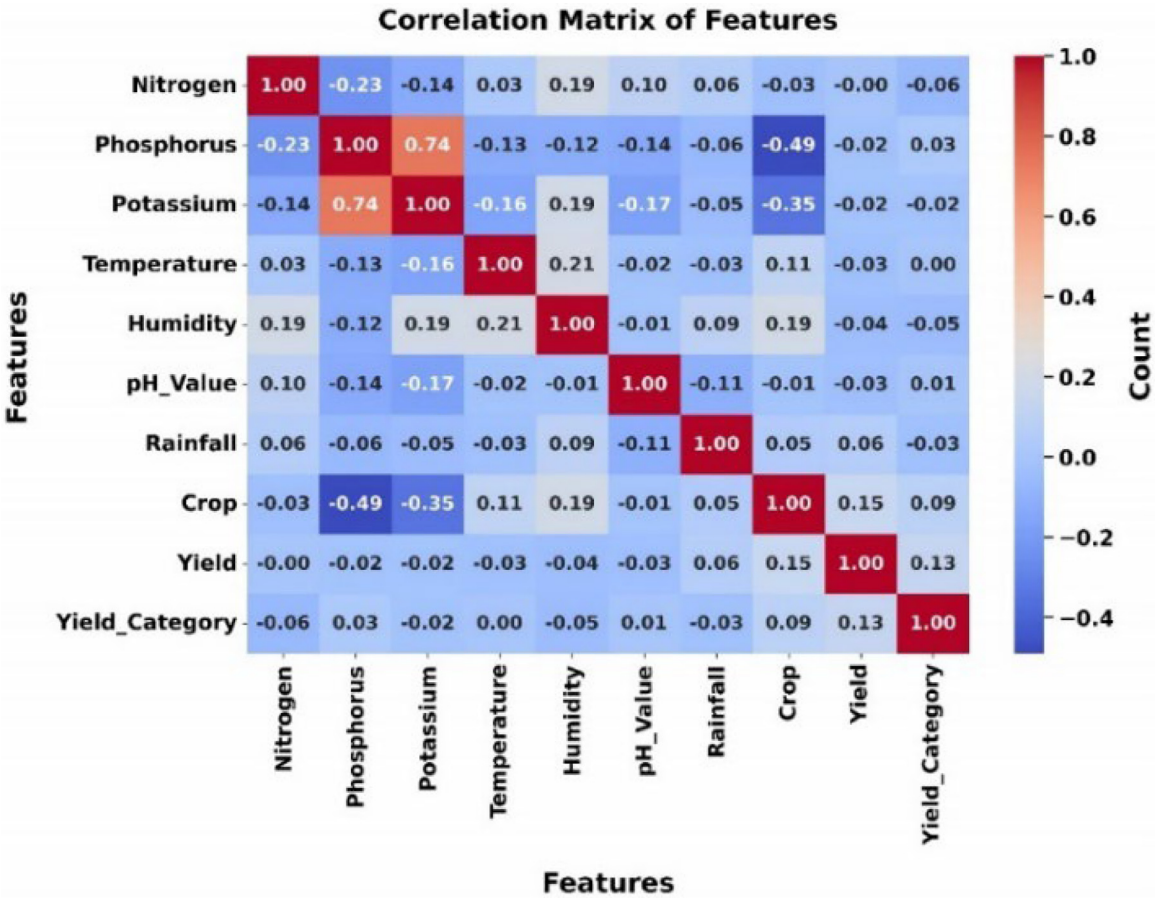

Fig. 6. Correlation analysis of soil nutrients and crop yield.

In Fig. 9, you can see the training and validation accuracies of an ML model used for CYP. The training accuracy values span from 0.5 to 1.0, as the model's fit to training dataset continually improves. As training goes on, the accuracies have continually increased from 0.5. They attain a peak of 1.0, indicating effective learning and further regression adaptation. However, the validation accuracy values display this growth with much greater diversity. Starting at 0.56, they fluctuate considerably before finally coming to a crest at highest levels on the chart of 1.0. This discrepancy between training and validation accuracies points to the potential for overfitting at certain stages with this model. Although good performance is achieved on training data, that consistency breaks down during validation round which bears heavily on all kinds of other activities. This analysis underlines the need to carefully tune model parameters such that it can generalize well and further highlights the necessity for the appropriate trade-off between complexity and performance in developing robust prediction models for SA (Fig. 10).

In the effort to build any machine learning models of accuracy or accuracy, therefore, this work utilizes regression to predict average corn output per field over a given span spanning three years.The training loss begins at 2.0 and gradually drops to 0.0, indicating in effect learning of actual characteristics from the training dataset.And you see all of this leaving room for your model to kick its heels in the air before settling into its final ohms, in which state you now start your "computation" for prediction set performance evaluation accordingly. The downturn demonstrates that where relationships with fellow training data were quite explicit of itself. The model has begun to impose some of structure imposing upon training points with greater than before result. The new instructional loss hence has a much more variable time path: It begins at 1.5, slows down to 0.3, and then settles at 0.0The VL decrease thus specifies that the model can generalize reasonably well from observed details to completely new cases with a little variation, implying itself as it captures major relationships in our validation.

Fig. 11 displays the classification metrics presented Meticulousness, Reminiscence, and F1 Score all achieving a value of 1.0 indicating a perfect performance of the ML model in classifying crop yield categories. Correctness measures the correctness of positive estimates, signifying that every instance classified as a positive yield category is indeed correct. A fastidiousness of 1.0 suggests no FP, meaning all predicted high-yield cases accurately reflect actual high yields. Recall, conversely, assesses the representation's facility to recognize all appropriate instances, affirming that it successfully recognizes every actual positive yield case without missing any, as evidenced by a perfect score. The F1 Score, which harmonizes strictness and reminiscence into a distinct metric, further confirms the model's optimal performance, particularly in circumstances where lesson balance is crucial. Collectively, these metrics demonstrate

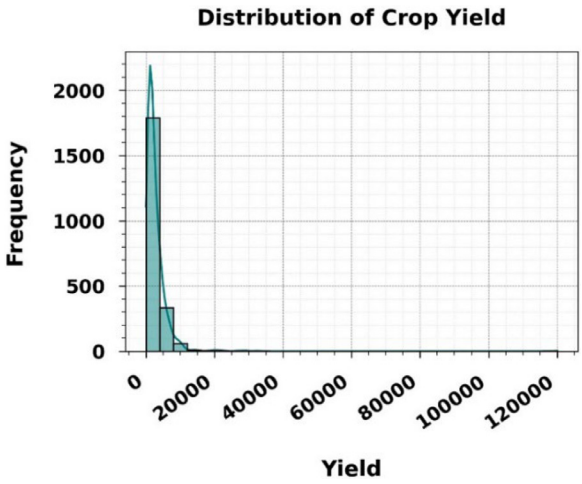

Fig. 7. Distribution of crop yield frequency values.

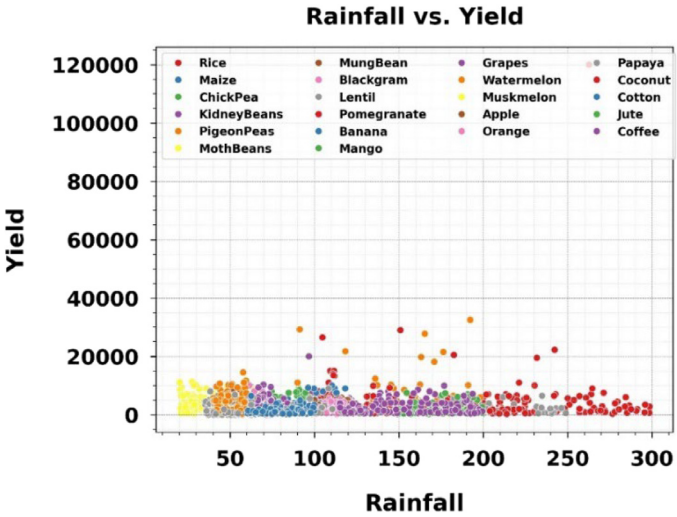

Fig. 8. Comparison of rainfall vs. yield crops.

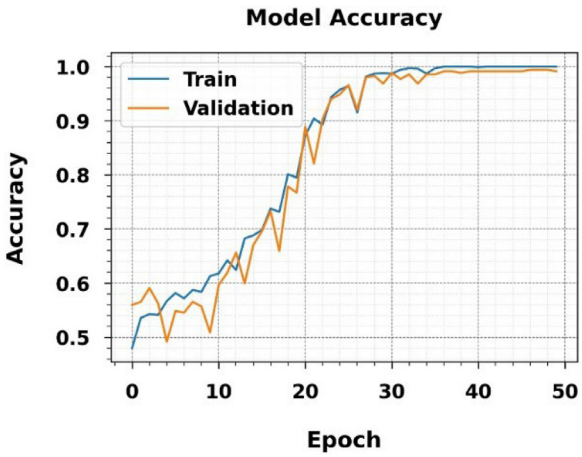

Fig. 9. Model accuracy in crop yield prediction.

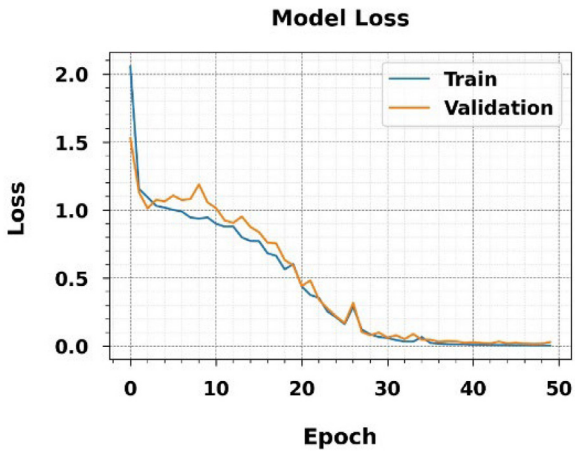

Fig. 10. Model loss in crop yield prediction.

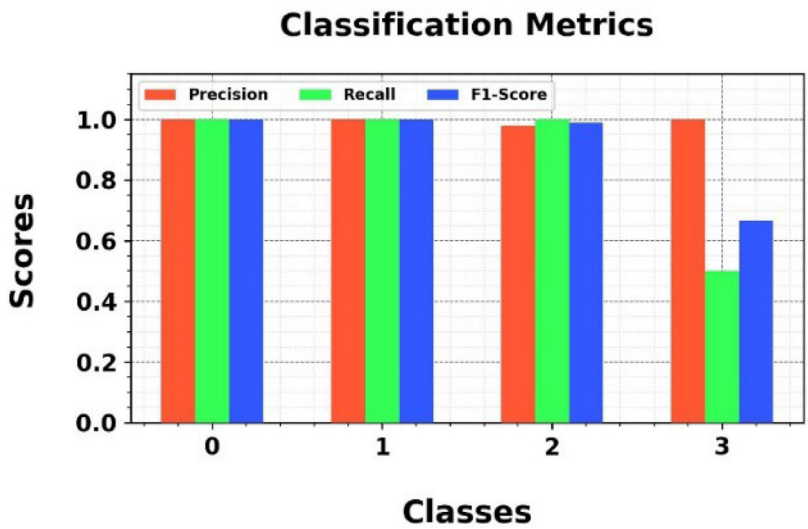

Fig. 11. Classification metrics precision, recall, and F1 score achieving 1.0.

Table 2
Performance comparison of different models for crop yield and soil health assessment.

Metric	CNN	RNN	RBFN	Proposed
Crop yield accuracy	75 %	95 %	80 %	97 %
Soil health assessment accuracy	65 %	90 %	85 %	92 %
Precision (yield)	70 %	90 %	78 %	93 %
Recall (yield)	70 %	90 %	76 %	94 %
F1-score (yield)	70 %	90 %	77 %	93 %

the model's effectiveness in enhancing CYP, playing a vivacious part in SA by allowing for informed decision-making and supply management.

A comparison of CNN, RNN, RBFN, and a suggested model's performance in terms of several metrics for predicting crop production and evaluating soil health is shown in Table 2. In every metric, the suggested model performs better than any alternative approach. With a 97 % accuracy rate for crop yield, the suggested model outperforms CNN (75 %), RNN (95 %), and RBFN (80 %). Comparing the suggested model to CNN, RNN, and RBFN, the accuracy of soil health evaluation increases to 92 % from 65 %, 90 %, and 85 %, respectively. In comparison to the lower values of CNN, RNN, and RBFN, the suggested model also shows a significant improvement in precision, recall, and F1-Score for crop yield, reaching 93 %, 94 %, and 93 %, respectively. When compared to conventional techniques, these data demonstrate how well the suggested model predicts crop production and soil health.

Conclusion

The research concludes that ML significantly enhances the accuracy and dependability of CYP and soil health assessment. By leveraging diverse data foundations consisting of remote sensing, soil sensors, and climatic data, ML models provide precise understandings that support agriculturalists make conversant decisions. The integration of techniques like iterative partitioning, back-propagation neural networks, and data fusion with geographical information systems has proven effective in improving predictive performance metrics, for occurrence accurateness, perfection, remembrance, and F1-score. These models enable early detection of soil degradation, optimize reserve use, and support SFP. The results demonstrate that ML can optimize agricultural productivity, lessen environmental impression, and promote long-term agricultural sustainability. Future improvements in real-time statistics integration and AI-enabled systems hold the potential to further enhance crop prediction accuracy and SHM for global agricultural development.

Limitations

None.

Ethics statements

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Kapil Netaji Vhatkar: Methodology, Software, Conceptualization, Writing – original draft. **Shweta Ashish Koparde:** Data curation, Methodology, Visualization, Writing – original draft. **Sonali Kothari:** Project administration, Methodology, Writing – review & editing, Funding acquisition. **Jayesh Sarwade:** Methodology, Visualization, Writing – review & editing. **Kishor Sakur:** Methodology, Conceptualization, Writing – review & editing.

Data availability

Data will be made available on request.

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